



STARLING ELECTROFORGE

RESEARCH AND DEVELOPMENT DIVISION
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Instrumentation Amplifier Test Bench with Microcontroller-Based Differential Signal Generator

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1 Design Requirements and Specifications

This project aims to design, build, and validate an instrumentation amplifier test bench with a microcontroller-based differential signal generator. The work is divided into six main deliverables.

1.1 Microcontroller-Based Differential Signal Generator

- Generate a 60 Hz sinusoidal signal using filtered PWM output from an Arduino Uno.
- Generate a 1 kHz sinusoidal signal using filtered PWM output from an Arduino Uno.
- Combine both sinusoidal signals to produce a noisy differential signal.
- Build and validate the circuit on a breadboard.
- Validate the quality of the sinusoidal output signals.

1.2 Three-Op-Amp Instrumentation Amplifier

- Design a three-op-amp instrumentation amplifier circuit.
- Calculate the required passive component values using 5% tolerance components.
- Select appropriate operational amplifiers.
- Build the circuit on a breadboard.
- Validate common-mode rejection performance through CMRR measurements.

1.3 Breadboard Characterization

- Measure the main circuit parameters of the breadboard implementation, including cutoff frequencies and gains.
- Measure the actual values of the passive components used in the noisy signal generator and instrumentation amplifier.
- Refine component values through iterative SPICE simulation and breadboard validation to achieve the desired performance.
- Measure the common-mode rejection ratio (CMRR) using 5% tolerance components by determining both differential gain and common-mode gain.



1.4 GitHub Repository and Documentation

- Create and maintain a GitHub repository for the project.
- Write a complete README file.
- Include Altium schematics.
- Include a design report documenting calculations, implementation, and validation results.
- Include visual support to illustrate signals at each filtering/amplifying stage.
- Include oscilloscope captures illustrating signal quality and behavior at each stage of the system.

1.5 Performance Improvement with Precision Components

- Replace critical passive components with 1% precision parts.
- Measure the common-mode rejection ratio (CMRR) using 1% components.
- Compare the results obtained with 5% and 1% resistor tolerances.

1.6 PCB Implementation

- Design and manufacture two 100 mm $\times$ 100 mm PCBs:
 - one PCB for the signal generator;
 - one PCB for the instrumentation amplifier.
- Add test points to validate/study the behavior of each filter and amplifier stage.
- Include proper connectors and terminal labels.